20.03.26 list datastructure

In [1]:
```
l=[]
1
```

Out[1]: 1

In [41]:
```
type(l)
```

Out[41]: list

In [42]:
```
l.append(20)
l.append(10)
l.append(30)
l.append(40)
print(l)
```

```
[10, 10, 10, 20, 10, 30, 40, 20, 10, 30, 40, 20, 10, 30, 40, 20, 10, 30, 40, 20,
10, 30, 40, 20, 10, 30, 40, 20, 10, 30, 40, 20, 10, 30, 40, 10, 10, 10, 20, 10, 3
0, 40, 20, 10, 30, 40, 20, 10, 30, 40]
```

In [45]:
```
l=[]   #check differences between these two
l.append(10)
l.append(20)
l.append(30)
l.append(40)
print(l)
```

```
[10, 20, 30, 40]
```

In [46]:
```
l2=[50,60,70,80]
l2
```

Out[46]: [50, 60, 70, 80]

In [47]:
```
print(id(l))
```

```
2037471243904
```

In [23]:
```
len(l2)
```

Out[23]: 4

In [24]:
```
l2.clear()
```

In [25]:
```
len(l2)
```

Out[25]: 0

In [27]:
```
l2
```

Out[27]: []

In [28]:
```
del l2
```

In [29]:
```
l2
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[29], line 1
----> 1 l2

NameError: name 'l2' is not defined
```

In [48]:
```
l.count(10) # count give no of time appeared
```

Out[48]:
```
1
```

In [34]:
```
a
```

Out[34]:
```
[1, 2, 3, 4, 5, 6]
```

In [53]:
```
l1=[10, 2.3, 1+2j, True, 'nit', [1,2,3]]
l1
```

Out[53]:
```
[10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]
```

In [54]:
```
len(l1)
```

Out[54]:
```
6
```

In [55]:
```
print(l)
print(l1)
```

```
[10, 20, 30, 40]
[10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]
```

In [56]:
```
print(id(l))
print(id(l1))
```

```
2037471243904
2037471696768
```

In [57]:
```
l2=l.copy()
l2
```

Out[57]:
```
[10, 20, 30, 40]
```

In [58]:
```
print(id(l)) # while argument or parameters are assigned to l2 of l, yet ids of
print(id(l1))
print(id(l2))
```

```
2037471243904
2037471696768
2037471459008
```

In [59]:
```
l2
```

Out[59]:
```
[10, 20, 30, 40]
```

In [60]:
```
len(l2)
```

Out[60]:
```
4
```

In [61]:
```
l2.clear()
```

In [62]:
```
l2
```

Out[62]:
```
[]
```

In [63]:
```
len(l2)
```

Out[63]:
```
0
```

In [64]:
```
l
```

Out[64]:
```
[10, 20, 30, 40]
```

In [65]:
```
l.count(10)  #count function gives no of times value appeared
```

Out[65]:
```
1
```

In [66]:
```
l.count(50) # here 50 is not in the list
```

Out[66]:
```
0
```

In [67]:
```
print(l)
print(l1)
```

```
[10, 20, 30, 40]
[10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]
```

In [70]:
```
a=[1,2,3] # you cant pass two argument
b=[4,5,6]

a.extend(b)
```

In [71]:
```
a
```

Out[71]:
```
[1, 2, 3, 4, 5, 6]
```

In [72]:
```
l2=[]
```

In [73]:
```
l2
```

Out[73]:
```
[]
```

In [78]:
```
l2.extend(l)
```

In [79]:
```
l2
```

Out[79]:
```
[10, 2.3, (1+2j), True, 'nit', [1, 2, 3], 10, 20, 30, 40, 10, 20, 30, 40]
```

In [80]:
```
l
```

Out[80]:
```
[10, 20, 30, 40]
```

In [81]:
```
l[0]
```

Out[81]: 10

In [84]:
```
l[0]=100
```

In [85]:
```
l[0]
```

Out[85]: 100

In [86]:
```
l
```

Out[86]: [100, 20, 30, 40]

In [87]:
```
l[-2]=300
l
```

Out[87]: [100, 20, 300, 40]

In [88]:
```
l.index(100)
```

Out[88]: 0

In [89]:
```
l2
```

Out[89]: [10, 2.3, (1+2j), True, 'nit', [1, 2, 3], 10, 20, 30, 40, 10, 20, 30, 40]

In [90]:
```
l2.index(10) # here 10 is three times but index always gives start index value
```

Out[90]: 0

In [91]:
```
l
```

Out[91]: [100, 20, 300, 40]

In [92]:
```
l.insert(2,30)
```

In [93]:
```
l
```

Out[93]: [100, 20, 30, 300, 40]

In [94]:
```
l2
```

Out[94]: [10, 2.3, (1+2j), True, 'nit', [1, 2, 3], 10, 20, 30, 40, 10, 20, 30, 40]

In [95]:
```
l2.pop() # this is default pop and it delets last value
```

Out[95]: 40

In [96]:
```
l2
```

Out[96]: [10, 2.3, (1+2j), True, 'nit', [1, 2, 3], 10, 20, 30, 40, 10, 20, 30]

In [97]:
```
l2.pop(2)  # here argument  is 2 and it will delete 2 index's value
```

Out[97]: (1+2j)

In [98]:
```
l2
```

Out[98]:
```
[10, 2.3, True, 'nit', [1, 2, 3], 10, 20, 30, 40, 10, 20, 30]
```

In [99]:
```
l2
```

Out[99]:
```
[10, 2.3, True, 'nit', [1, 2, 3], 10, 20, 30, 40, 10, 20, 30]
```

In [104…
```
l2.remove([1,2,3])
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[104], line 1
----> 1 l2.remove([1,2,3])

ValueError: list.remove(x): x not in list
```

In [101…
```
l2
```

Out[101…
```
[10, 2.3, True, 'nit', 10, 20, 30, 40, 10, 20, 30]
```

In [105…
```
l2.remove(2.3)  # so therefore the difference between pop and remove is pop dele
```

In [106…
```
l2
```

Out[106…
```
[10, True, 'nit', 10, 20, 30, 40, 10, 20, 30]
```

In [110…
```
l2.remove(True,'nit')
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[110], line 1
----> 1 l2.remove(True,     )

TypeError: list.remove() takes exactly one argument (2 given)
```

In [111…
```
l
```

Out[111…
```
[100, 20, 30, 300, 40]
```

In [112…
```
l.reverse()  # just reverse all the values as it is
```

In [113…
```
l
```

Out[113…
```
[40, 300, 30, 20, 100]
```

In [114…
```
l.sort() # it sorts the value in ascending order
```

In [115…
```
l
```

Out[115…
```
[20, 30, 40, 100, 300]
```

In [116…
```
l.sort(reverse=True) # this sorts reversed in descending order
```

In [117…
```
l
```

Out[117… [300, 100, 40, 30, 20]

In [118…
```
l3=[10,20,30,[2,3],40] # nested list, here we are replacing the entire index at
```

In [119…
```
l3[3]=45
```

In [120…
```
l3
```

Out[120… [10, 20, 30, 45, 40]

In [121…
```
l
```

Out[121… [300, 100, 40, 30, 20]

In [122…
```
500 in l
```

Out[122… False

In [123…
```
l
```

Out[123… [300, 100, 40, 30, 20]

In [125…
```
for i in l:
    print(i)
```

```
300
100
40
30
20
```

In [128…
```
for i in range(len(l)):
print(i, l[i])
```

```
  Cell In[128], line 2
    print(i, l[i])
    ^
IndentationError: expected an indented block after 'for' statement on line 1
```

In [ ]: